RESEARCH LETTER

Spectral-neighbourhood self-modulation for lightweight hyperspectral image classification with scarce labels

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ABSTRACT
Lightweight convolutional networks remain attractive for hyperspectral image (HSI) classification when labels are scarce. Their channel-attention blocks, however, usually derive the gating weights from global average pooling followed by fully connected layers, which ignores how statistics vary across neighbouring feature channels. We propose spectral-neighbourhood self-modulation (SNSM), a residual-block modulation in which each channel's scale and shift are predicted by a small one-dimensional convolution over three per-channel descriptors (mean, variance and cross-channel gradient). SNSM adds about 130 parameters per block and is initialised as the identity. Under one fixed protocol (five seeds, 100 epochs, no validation-based model selection), SNSM reaches 94.16±1.07% average accuracy (AA) on Indian Pines and 92.25±1.84% on Kennedy Space Center with 5% labelled pixels. This is the highest AA among the SE, ECA, CBAM, HybridSN and SSFTT baselines, and SNSM also leads in five of six settings across the 1%, 5% and 10% label budgets. Paired ablations attribute the gain mainly to the variance descriptor and the identity initialisation; the gradient descriptor and the additive shift each change AA by less than one point. Overall accuracy differs little between the convolutional variants. The AA gains concentrate in small classes, and on easier scenes all variants perform alike.

KEYWORDS
Hyperspectral image classification; channel attention; feature modulation; few labels; lightweight CNN

1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Kernel classifiers such as support vector machines (Camps-Valls and Bruzzone 2005; Melgani and Bruzzone 2004) and spectral–spatial methods (Fauvel et al. 2013; Ghamisi et al. 2017) were the standard before deep learning. Deep models now dominate (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019). Stacked autoencoders (Chen et al. 2014), recurrent networks (Mou, Ghamisi, and Zhu 2017) and graph convolutional networks (Hong et al. 2021) have all been applied. Convolutional neural networks (CNNs) that operate on spatial–spectral patches are still the usual lightweight choice (Hu et al.

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2015; Makantasis et al. 2015; Chen et al. 2016; Li, Zhang, and Shen 2017; Zhong et al. 2018; Roy et al. 2020). Transformers bring global context, but at a higher training cost (Hong et al. 2022; Sun et al. 2022).

Channel attention is a cheap way to make a CNN input-dependent. Squeeze-and-excitation (SE) (Hu, Shen, and Sun 2018) predicts one scale per channel from globally pooled features through two fully connected (FC) layers. CBAM adds spatial attention (Woo et al. 2018); ECA replaces the FC layers by a one-dimensional convolution over the pooled channel vector (Wang et al. 2020); SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019); GCT normalises per-channel statistics and gates them with $1+\tanh(\cdot)$ from an identity start (Yang et al. 2020); FcaNet replaces average pooling by several frequency components (Qin et al. 2021). All of these predict a scale only. FiLM adds a shift but takes its conditioning from an external input (Perez et al. 2018). Hyperspectral networks have adopted these blocks widely, usually combining SE-style channel gates with spatial attention (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021). With very few labelled pixels, FC gating has many parameters relative to the training set. Global means alone also discard how the response is spread within the patch.

We study a modulation that is (i) local across channels, (ii) driven by several per-channel statistics rather than the mean only, and (iii) identity at initialisation. Our contributions are:

- SNSM, a residual-block modulation whose scale and shift come from a dilated one-dimensional convolution over normalised channel means, variances and cross-channel gradients (Section 2);
- a single-protocol comparison on four scenes (plus two near-saturated scenes for SNSM alone), with label budgets of 1%, 5% and 10% on the two hardest scenes, in which every model, including HybridSN and SSFTT, is trained on the same splits, inputs and schedule with no validation-based checkpoint selection (Section 3);
- paired same-seed ablations that separate the effects of the descriptors, the identity initialisation and the additive shift (Section 4).

2. Method

2.1. *Input features*

Each band of the raw cube is smoothed with a 5×5 adaptive Wiener filter. Principal component analysis (PCA) (Jolliffe and Cadima 2016) then reduces the spectrum to 30 components; it is fitted without labels on all pixels of the scene. A one-level db4 discrete wavelet transform (Mallat 1989; Daubechies 1992) is applied to each component. Its approximation sub-band, resized bilinearly to the image size, is concatenated with the components, giving $B = 60$ features per pixel. Each labelled pixel is represented by the $15 \times 15 \times B$ patch centred on it (zero padding at the image border).

2.2. *Backbone*

A 3×3 convolution maps the B input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each

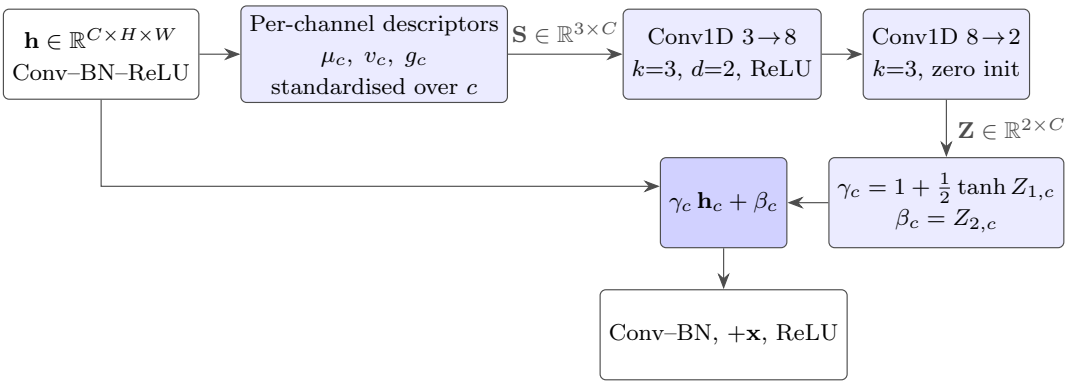


Figure 1. SNSM inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale γ_c and a shift β_c for each channel. The second convolution starts at zero, so the block is initially the identity.

computing

$$\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times 3}(\mathbf{x}))), \qquad \mathbf{y} = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}_{3\times 3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x}\big), \quad (1)$$

where $\mathcal{M}$ is the modulation under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores.

2.3. *Spectral-neighbourhood self-modulation*

For $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, SNSM first computes three descriptors for every channel c: the spatial mean μ_c, the spatial variance v_c, and the cross-channel gradient $g_c = (\mu_{c+1} - \mu_{c-1})/2$ (one-sided differences at the first and last channel). Each descriptor vector is standardised across channels, $\tilde{s} = (s - \bar{s})/(\sigma_s + \epsilon)$, and the three are stacked into $\mathbf{S} \in \mathbb{R}^{3\times C}$. Two one-dimensional convolutions along the channel axis give

$$\mathbf{Z} = \mathrm{Conv1D}^{(2)}_{k=3}\big(\mathrm{ReLU}(\mathrm{Conv1D}^{(1)}_{k=3,\,d=2}(\mathbf{S}))\big) \in \mathbb{R}^{2\times C}, \quad (2)$$

with 8 hidden channels, dilation $d = 2$ in the first layer and replicate padding. The modulation is

$$\gamma_c = 1 + \tfrac{1}{2}\tanh(Z_{1,c}), \qquad \beta_c = Z_{2,c}, \qquad \mathcal{M}(\mathbf{h})_c = \gamma_c\,\mathbf{h}_c + \beta_c. \quad (3)$$

The weights and bias of $\mathrm{Conv1D}^{(2)}$ are initialised to zero, so every block starts as the plain residual block and $\gamma_c \in [0.5, 1.5]$ thereafter. The modulation adds 130 parameters per block, against 2,128 for SE gating with reduction to 16 units. Each channel's parameters depend on a window of neighbouring channels (receptive field of seven channels), as in ECA (Wang et al. 2020). Unlike ECA, SNSM uses dispersion and gradient descriptors in addition to the mean, outputs a shift as well as a scale, and starts from the identity. The bounded gate and identity start resemble GCT (Yang et al. 2020), and SRM also uses channel dispersion (Lee, Kim, and Nam 2019). SNSM differs from both in filtering several descriptors jointly across neighbouring channels and in predicting a shift. The SNSM (γ) variant drops the shift output. Figure 1 shows the data flow.

Table 1. Benchmark scenes and the 5% splits used (seed 42; split sizes are the same for every seed).

Scene	Sensor	Size (pixels)	Bands	GSD (m)	Classes	Labelled	Train/test
Indian Pines (IP)	AVIRIS	145×145	200	20	16	10,249	505/9,744
Pavia University (PU)	ROSIS	610×340	103	1.3	9	42,776	2,135/40,641
Kennedy Space Center (KSC)	AVIRIS	512×614	176	18	13	5,211	256/4,955
Botswana (BW)	Hyperion	1476×256	145	30	14	3,248	156/3,092
Salinas	AVIRIS	512×217	204	3.7	16	54,129	2,697/50,000
WHU-Hi HanChuan	Headwall Nano-Hyperspec	1217×303	274	0.109	16	257,530	12,868/50,000

3. Experimental setup

Data. We use Indian Pines (IP), Pavia University (PU), Kennedy Space Center (KSC), Botswana (BW), Salinas and WHU-Hi HanChuan (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020; Zhong et al. 2020). For each class, 5% of the labelled pixels (at least one) are drawn at random for training; the remaining labelled pixels are used for testing. For Salinas and WHU-Hi, a random subset of 50,000 test pixels is used. This gives 505/9,744 (IP), 256/4,955 (KSC), 2,135/40,641 (PU) and 156/3,092 (BW) training/test pixels (Table 1). At 5%, three IP classes (Alfalfa, Grass-pasture-mowed and Oats) have only one or two training pixels.

Models. On IP, KSC, PU and BW we compare seven modulation variants on the same backbone, plus HybridSN (Roy et al. 2020) in its original layer configuration. The variants are: plain CNN; SE (γ) (Hu, Shen, and Sun 2018); ECA (γ) (Wang et al. 2020), with kernel size 3 as given by its adaptive rule for 64 channels; CBAM (Woo et al. 2018), with channel reduction 16 and a 7×7 spatial gate; FC (γ+β), i.e. SE-style FC layers predicting a sigmoid scale and a linear shift; SNSM (γ); and SNSM (γ+β). SSFTT (Sun et al. 2022) (four tokens, one transformer layer with eight heads) is reimplemented and run on IP and KSC only. Salinas and WHU-Hi are near saturation and are run with SNSM only.

Protocol. Every model sees the same 60-feature patches and the same splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate 10^{-3}, weight decay 10^{-4}), a step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, with no data augmentation and no early stopping. The final-epoch model is evaluated once. No validation or test data are used for model selection. Each configuration is repeated with seeds 42–46, and we report mean $\pm$ sample standard deviation. Runs used PyTorch (Paszke et al. 2019) on an NVIDIA Tesla T4. Baselines are therefore trained under our protocol and inputs rather than their original settings, so their accuracies are not directly comparable to published values.

4. Results

Accuracy. Table 2 gives the main comparison. On IP, SNSM (γ+β) has the highest AA; SSFTT and CBAM have higher OA but lower AA. On KSC, SNSM (γ+β) has the highest OA and AA, and its seed-to-seed spread is several times smaller than that of the plain and SE models. On PU and BW the seven CNN variants are within a few tenths of a point of each other, so modulation brings no measurable benefit on these scenes. HybridSN trained under our protocol is less accurate and less stable; on one BW seed it did not converge (9.4% OA). SNSM reaches 99.90$\pm$0.04% OA on Salinas and 99.64$\pm$0.11% OA on WHU-Hi. Figure 2 shows the AA spread across seeds on the

Table 2. OA and AA (%, mean ± std over five seeds) at 5% labelled pixels. Best mean per column in bold.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	95.96±0.89	87.86±1.29	90.93±6.94	85.64±12.98	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE (γ)	96.23±0.69	89.49±2.39	91.54±5.56	87.32±8.36	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA (γ)	96.17±0.74	91.30±2.26	88.43±6.66	83.72±9.93	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.50±0.48	92.87±1.14	92.12±1.57	89.00±1.82	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
FC (γ+β)	96.24±0.52	92.51±1.60	84.31±6.34	77.25±8.18	99.51±0.06	98.88±0.14	98.80±0.70	98.63±0.98
SNSM (γ)	96.19±0.74	93.70±1.38	94.49±2.02	91.63±3.04	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.29±0.74	**94.16±1.07**	**94.55±1.25**	**92.25±1.84**	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03
HybridSN	88.30±4.91	73.64±8.28	68.04±9.32	55.19±11.33	97.84±1.52	95.46±4.12	77.16±37.92	75.56±38.31
SSFTT	**96.71±0.79**	92.35±1.81	80.44±2.57	68.50±3.65	–	–	–	–

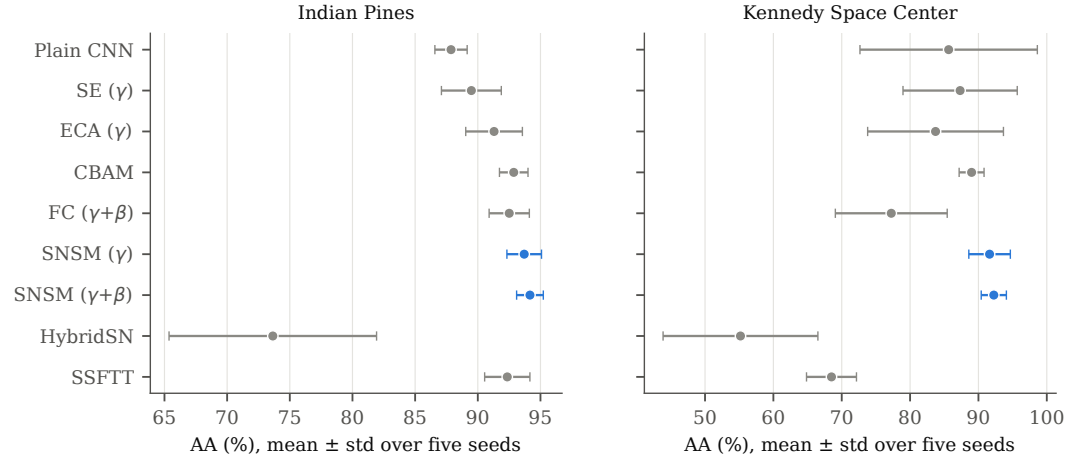


Figure 2. AA (mean ± standard deviation over five seeds) on Indian Pines and Kennedy Space Center. SNSM variants in blue.

two hard scenes.

Where the gain comes from. Table 3 and Figure 3 pair runs that share a seed and therefore a split. Replacing SE gating by the SNSM descriptor network (both scale-only) raises AA on IP and KSC in nine of ten paired runs, with mean gains above 4 points. ECA also gates each channel with a local convolution across channels, but it uses only the channel mean and has lower AA than SNSM (γ) on IP (four of five seeds) and KSC (three of five seeds, with a large seed-to-seed spread for ECA). Locality across channels alone therefore does not explain the gain; Table 4 separates the descriptors and the initialisation. CBAM, which adds max pooling and a spatial gate to SE-style channel attention, also has lower AA than SNSM (γ+β) on IP (four of five seeds) and KSC (five of five seeds). Adding the shift to SNSM changes AA by less than one point and helps in only three of five seeds on either scene, so we do not claim a benefit from the shift itself. Adding a shift to FC gating helps on IP but lowers AA by about 10 points on KSC. This suggests that an unconstrained FC shift is unstable with 256 training pixels; SNSM's bounded scale and zero-initialised output avoid this problem.

Descriptors and initialisation. Table 4 removes descriptors from SNSM (γ+β) or drops its zero initialisation, again pairing runs by seed. The variance descriptor carries most of the gain: with the mean alone, AA falls by 1.16 points on IP and 8.44 points on KSC, and adding the gradient to the mean does not recover it. Once the variance is present, the gradient changes AA by less than one point (it lowers AA by 0.39 points on IP and raises it by 0.92 on KSC). Without zero initialisation AA is lower in all ten paired runs, by 1.49 points on IP and 1.91 on KSC.

Table 3. Paired same-seed differences in AA (percentage points): mean over five seeds and number of seeds with a positive difference.

ΔAA (wins/5)	IP	KSC	PU	BW
SNSM (γ+β) − SNSM (γ)	+0.46 (3/5)	+0.62 (3/5)	+0.02 (3/5)	-0.03 (1/5)
FC (γ+β) − SE (γ)	+3.02 (5/5)	-10.07 (0/5)	+0.09 (1/5)	-0.13 (1/5)
SNSM (γ) − SE (γ)	+4.22 (5/5)	+4.31 (4/5)	+0.11 (1/5)	+0.13 (2/5)
SNSM (γ) − ECA (γ)	+2.41 (4/5)	+7.91 (3/5)	-0.09 (2/5)	-0.02 (2/5)
SNSM (γ+β) − CBAM	+1.30 (4/5)	+3.25 (5/5)	-0.18 (1/5)	+0.07 (4/5)
SNSM (γ+β) − Plain CNN	+6.30 (5/5)	+6.60 (4/5)	-0.08 (2/5)	-0.01 (2/5)

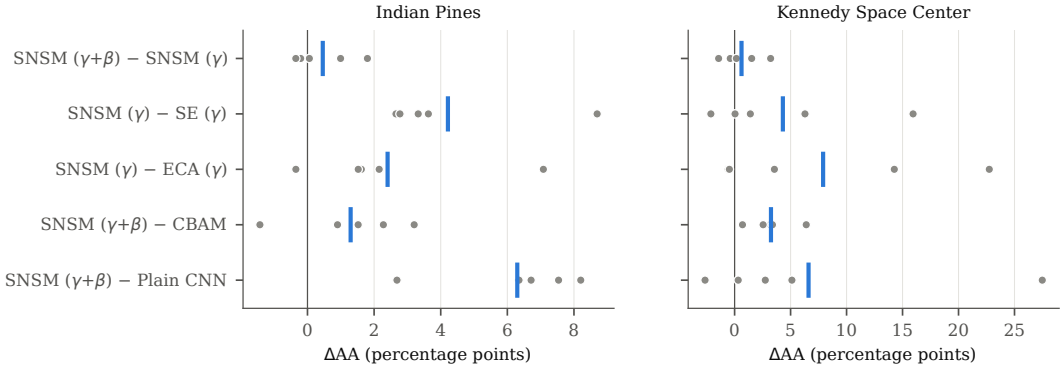


Figure 3. Paired same-seed differences in AA. Grey dots are the five seeds; the blue bar is their mean.

Label budget. Table 5 repeats the comparison with 1% and 10% of the labelled pixels per class on IP and KSC (98 and 48 training pixels at 1%; 1,018 and 516 at 10%). SNSM (γ+β) has the highest AA in five of the six settings; at 10% on KSC, CBAM is higher. On IP at 1% and 10% its margin over the CNN baselines is 0.4–2.0 points. At 1% on IP it also has the lowest OA of the CNN variants (82.77% against 84.23–85.26%), so its advantage is again in AA rather than OA. On KSC at 10%, every model is unstable across seeds: on seed 42 all models reach between 52% and 77% AA, and every model has a lower mean AA than at 5%. We have not identified the cause, so this setting should be read with caution.

The AA gains on IP come from small classes (Table 6). Relative to the plain CNN, SNSM (γ+β) raises mean per-class accuracy on Oats (20 labelled pixels) from 21.1% to 74.7%, on Grass-pasture-mowed (28) from 63.0% to 92.6%, and on Stone-Steel-Towers (93) from 76.9% to 92.4%. On KSC the largest change is for Cabbage palm hammock, from 60.2% to 93.1%. Oats and Grass-pasture-mowed have one training pixel and 19 and 27 test pixels, so these two gains correspond to about ten and eight test pixels. AA on IP is therefore sensitive to a few pixels, whereas the OA of all seven CNN variants on IP lies within 0.6 points. Figure 4 shows the resulting IP classification map for one seed.

Cost. SNSM adds almost no parameters or multiply–accumulates to the plain CNN (Table 7), but its descriptor computation consists of many small operations, so its batch-1 latency is about 3.7 times and its training time about 1.7 times that of the plain CNN. SNSM is lightweight in parameters and multiply–accumulates, not in wall-clock time. All CNN variants remain an order of magnitude smaller than HybridSN.

Table 4. Descriptor and initialisation ablation of SNSM (γ+β) at 5% labelled pixels. AA in % (mean ± std over five seeds); ΔAA is the paired difference from the full model and wins counts seeds on which the variant is better.

Descriptors	IP		KSC	
	AA	ΔAA (wins/5)	AA	ΔAA (wins/5)
mean, var, grad (SNSM)	94.16±1.07	–	92.25±1.84	–
mean	93.01±0.72	-1.16 (1/5)	83.81±7.78	-8.44 (0/5)
mean, var	94.55±1.67	+0.39 (4/5)	91.33±2.43	-0.92 (1/5)
mean, grad	93.37±1.09	-0.80 (1/5)	82.35±8.54	-9.90 (1/5)
mean, var, grad; default init.	92.67±1.15	-1.49 (0/5)	90.34±1.27	-1.91 (0/5)

Table 5. AA (%, mean ± std over five seeds) with 1%, 5% and 10% labelled pixels per class. The 5% column repeats Table 2. Best mean per column in bold.

Method	IP			KSC		
	1%	5%	10%	1%	5%	10%
Plain CNN	83.51±1.23	87.86±1.29	96.69±1.43	55.40±2.45	85.64±12.98	76.90±10.69
SE (γ)	84.01±1.66	89.49±2.39	96.42±2.04	56.05±5.00	87.32±8.36	78.84±12.82
ECA (γ)	84.66±1.20	91.30±2.26	96.79±1.42	54.76±3.84	83.72±9.93	77.85±8.42
CBAM	83.08±1.15	92.87±1.14	96.20±1.94	60.47±2.79	89.00±1.82	**87.12±8.78**
SNSM (γ+β)	**85.04±0.76**	**94.16±1.07**	**97.70±0.66**	**61.83±4.57**	**92.25±1.84**	85.44±14.47
SSFTT	80.68±1.51	92.35±1.81	96.73±1.06	54.03±5.02	68.50±3.65	60.18±9.86

5. Discussion and limitations

Local cross-channel modulation helps where it is most needed: scenes with small, spectrally similar classes (IP, KSC). It does so with fewer gating parameters than SE. On scenes where a plain CNN is already near saturation, its benefit disappears. Several limitations apply. Training and test pixels are drawn at random from the same scene, so overlapping patches share spatial context and accuracies are optimistic compared with spatially disjoint splits (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). PCA is fitted on all pixels of each scene without labels. Five seeds give only an indication of variability, and the paired differences in Table 3 are not formal significance tests (with five pairs, a two-sided Wilcoxon signed-rank test cannot reach $p < 0.05$). Baselines were not tuned to our protocol. HybridSN and SSFTT are far less accurate here than in their original papers, which used larger training sets and their own preprocessing and schedules, so our results describe only how they behave under this shared protocol. SNSM's hidden width, kernel size, dilation and scale bound were fixed in earlier exploratory runs on these scenes and were not tuned further. We report no sensitivity analysis for them. Results on KSC at 10% labels vary strongly across seeds for every model. Finally, the channel order of learned feature maps has no physical wavelength meaning, so "neighbourhood" refers to adjacency of feature channels, which the first convolution is free to learn.

6. Conclusion

SNSM predicts per-channel scale and shift with a small convolution over neighbouring channels' mean, variance and gradient. Under a single strict protocol, with about 130 extra parameters per block, it gives the best average accuracy on Indian Pines and KSC at 5% labels, and in five of six settings across the 1%, 5% and 10% label budgets. Paired ablations attribute the improvement mainly to the variance descriptor and the identity initialisation, not to the gradient descriptor or the additive shift. The gains

Table 6. Per-class accuracy (%, mean over five seeds) on Indian Pines; n is the number of labelled pixels. Best value per row in bold.

Class	n	Plain CNN	SE (γ)	ECA (γ)	CBAM	SNSM (γ)	SNSM (γ+β)	HybridSN	SSFTT
Alfalfa	46	92.7	90.9	90.5	92.7	**94.5**	89.1	29.1	81.8
Corn-notill	1428	96.3	95.8	96.7	96.5	96.6	95.8	88.1	**97.3**
Corn-mintill	830	93.9	94.0	93.6	**95.0**	93.7	94.2	82.2	94.4
Corn	237	90.9	92.1	92.8	94.3	93.8	**96.5**	61.8	90.6
Grass-pasture	483	97.1	**97.2**	96.5	96.8	96.7	96.9	82.3	96.3
Grass-trees	730	97.3	96.5	97.2	97.8	97.2	97.7	97.8	**98.4**
Grass-pasture-mowed	28	63.0	58.5	84.4	89.6	88.9	**92.6**	44.4	**92.6**
Hay-windrowed	478	99.3	**99.6**	99.6	99.6	99.4	99.6	96.3	99.5
Oats	20	21.1	44.2	43.2	58.9	70.5	**74.7**	27.4	60.0
Soybean-notill	972	92.3	93.7	92.3	93.3	93.2	92.7	84.5	**94.2**
Soybean-mintill	2455	97.3	**97.9**	97.2	97.1	96.7	97.2	94.4	97.4
Soybean-clean	593	93.1	92.8	92.4	93.8	91.8	91.2	73.8	**94.5**
Wheat	205	96.4	97.7	96.6	97.1	96.8	98.2	85.4	**98.6**
Woods	1265	99.5	99.6	99.6	99.5	99.4	99.2	96.8	**99.8**
Bldg-Grass-Trees-Drives	386	**98.9**	98.1	98.8	98.2	98.1	98.7	89.1	97.9
Stone-Steel-Towers	93	76.9	83.1	89.4	85.6	91.7	**92.4**	44.9	84.3
AA		87.9	89.5	91.3	92.9	93.7	94.2	73.6	92.4

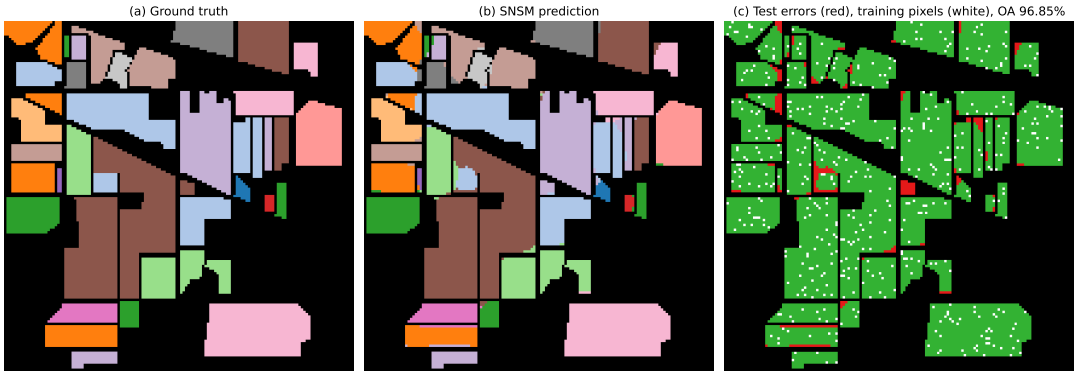


Figure 4. Indian Pines, SNSM (γ+β), seed 42: (a) ground truth, (b) prediction for all labelled pixels, (c) test errors in red and training pixels in white.

are in average rather than overall accuracy, and they cost a 3.7-fold increase in batch-1 latency.

Disclosure statement

The authors report there are no competing interests to declare.

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Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020; Zhong et al. 2020). Code, per-run results and the scripts that generate every table are available at `https://github.com/Devguru-codes/research_paper_code`.

Table 7. Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs).

Method	Params (M)	MMACs[a]	Latency (ms)[b]	Train IP (s)	Train PU (s)
Plain CNN	0.184	41.2	0.69	4.4	17.6
SE (γ)	0.188	41.2	1.17	4.6	19.3
ECA (γ)	0.184	41.2	0.98	5.2	19.8
CBAM	0.185	41.3	1.55	6.0	24.9
FC (γ+β)	0.190	41.2	1.31	4.7	19.8
SNSM (γ)	0.184	41.3	2.54	7.5	29.5
SNSM (γ+β)	0.184	41.3	2.54	7.6	30.2
HybridSN	1.743	137.4	0.71	26.2	115.3
SSFTT	0.291	34.9	1.25	7.6	–

[a] Multiply–accumulates of convolution and linear layers counted by `thop`; tensor operations such as attention products and descriptor statistics are not included. [b] Batch size 1 on a Tesla T4.

Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

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